

Customer Description

Organizations are increasingly deploying intelligent edge devices in industrial environments to perform sensing, analytics, automation, and AI inference close to where data is generated. These devices often operate in remote or untrusted environments where configuration files, software components, or AI models may be modified, replaced, or injected by unauthorized parties.

We want to develop **Edge Pico**, a secure Industrial IoT (IIoT) edge platform that establishes and maintains a **chain of trust** throughout the deployment lifecycle.

This project builds upon an existing **Configurable Edge Pico Platform** that supports modular services, deployable configurations, and AI workloads. The objective is not to redesign the platform itself, but to design and implement the security architecture required to establish a **chain of trust** across the deployment lifecycle, ensuring that only trusted configurations, software components, firmware images, and AI models can be deployed and executed on the device.

The platform shall ensure that every critical artifact deployed to the device, including configuration files, firmware components, software modules, and AI models, can be traced back to an authorized source and verified before use. Any asset that cannot be authenticated or whose integrity cannot be verified shall be rejected.

The device shall support a secure deployment workflow where only authorized personnel and approved systems can provision, configure, update, or manage deployed devices. The platform must protect against accidental corruption, malicious modification, unauthorized updates, and software supply-chain attacks.

The primary objective is not simply to secure communications, but to establish a verifiable **chain of trust** from the asset creator to the deployed edge device. Every stage of the lifecycle should contribute to maintaining confidence that:

- The sender is authorized.
- The asset originated from a trusted source.
- The asset has not been tampered with during transmission.
- The asset has not been modified while stored.
- The asset is approved for execution.
- The device can verify these properties before activation.

The platform should apply these trust guarantees consistently across:

- Configuration files
- Firmware images

- Software modules
- AI and machine learning models
- Security-sensitive operational commands

The completed solution should demonstrate a secure Industrial IoT deployment architecture where a **chain of trust** is maintained from development and provisioning through deployment and runtime execution.

Motivation

Industrial edge devices increasingly rely on configuration-driven applications, remote software updates, and deployable AI models.

In these environments, the most important question is often not:

"Does the AI model work?"

but rather:

"Can the device trust the AI model it is about to execute?"

Similarly:

- Can the device trust a new configuration file?
- Can it trust a remote update?
- Can it trust the entity attempting to perform the update?

A break anywhere in this process breaks the **chain of trust** and undermines the security of the entire deployment.

This project focuses on designing the mechanisms required to establish, maintain, and verify that chain of trust throughout the lifecycle of an Industrial IoT device.

Short Product Vision Statement

Develop a **secure IIoT edge platform that enforces a verifiable chain of trust**, ensuring that only authenticated, authorized, and integrity-verified configurations, software components, and AI models can be deployed and executed.

Key Security Concerns

Students should consider how the platform addresses:

Identity & Authorization

- How can the device verify who is attempting to deploy a configuration or update?
- How can unauthorized operators be prevented from managing the device?

Configuration Trust

- How can the device verify that a configuration file originated from an authorized source?
- How can tampering be detected before the configuration is applied?

Software Trust

- How can firmware and software modules be validated before loading or execution?
- How can unauthorized code be prevented from running?

AI Model Trust

- How can an AI model be verified before deployment?
- How can modifications to an approved model be detected?
- How can the device ensure it is executing the intended model version?

Lifecycle Trust

- How can trust be maintained throughout provisioning, deployment, updates, storage, and execution?
- How can audit records support accountability and traceability?

Requirements Students Can Infer

Area	Questions students should resolve
Chain of Trust Architecture	How is trust established and maintained throughout the asset lifecycle?
Authentication	How are operators, servers, and devices identified and verified?
Authorization	Who is allowed to deploy or modify assets?
Integrity Verification	How is tampering detected?

Configuration Management	How are trusted configurations delivered and validated?
AI Model Verification	How are models authenticated before loading?
Firmware Validation	How is software integrity protected?
Key Management	How are trust anchors and cryptographic secrets managed?
Auditability	How are trust decisions recorded and traced?
Recovery	What should happen when verification fails?
Deployment Workflow	How does trust flow from developer to deployed device?
Scalability	How can the architecture support large industrial deployments?